④ Diagnostic Evaluation

↳ no training loops ⇒ No opti. updates, No Adam / L-BFGS
No training loss calculation, we are inverting existing model

↳ Load trained LBFGS & Adam

Understand every code before moving to 20 Nov. storing

(#) why not RAR-G directly cuz we won't know what RAR-G fixed.

↳ Dense Grid => 256 spatial co-ordinates & 100 time co

~~258~~ 256 × 100 = 25,600
↳ from burgers.npz

↳ Calc. PDE res.

$$\boxed{R_\theta = u_t + u_\theta u_x - \nu u_{xx}}$$

↳ using autodiff. → (2×10)

(·) Calc. odd symmetry errors:

↳ evaluate model at $\boxed{(t, x) \ \& \ (t, -x)}$

$$u_\theta(-x, t) + u_\theta(x, t)$$

PDE plot =)

(i) x vs Residual map |R|

(Contr. PDE-Resi)

Max |R| => 1.928921×10^{2}

R mean => 4.611992×10^{-1}

Max Sym err => 2.157272 ~~×10~~

Mean Sym error => 4.83986×10^{-1}

deri. calc. check

Date: / /

During BZ training ⇒ ① L-BFGS loss ⇒ 4×10^{-5} at 2000 itr.

② Diagnostic summary ⇒ Max PDE Res: ⇒ 192.89

(a) LBFGS perf. curve-fit 2000 itr it was given, driving error to zero exactly at some co-ordinates, but allowed eqns to violently explode in between.

↳ The bright streak in PL traces the path of missing shockwave, where physics were violated.

↳ (i) Need to verify deri. calcu. & checkpt before interpreting

② $\boxed{t \text{ vs } x \text{ vs symm. error } |u(x) + u(-x)|}$ these values as PDE gen violates

(odd symm violation)

Ini condi. ⇒ $u(x,0) = -\sin(\pi x)$ → odd fn

$u(-x) = -u(x)$

Burgers → preserve this symm. → (ie) shock at $x=0$ is needed

· Max symm error ⇒ 2.19

Mean abs error → 0.484 ⇒ $x \simeq 0.45$

MATRIKAS

③ t vs x vs Abs err. vs ref. Ref. errors

↳ Network isn't inacc. everywhere → in black region

↳ error is mostly conc. in b/w the $x=0$ to $x=0.45$ (yellow/white/ bands)

white shows should be

↳ in PINN but

↳ so we need to fix this gap

↳ other reg. are highly acc. (black region)

④ Symm. Violation expl. ~~(Ref. errors)~~

$$\underline{u_1+u_2=0}$$

for symm ⇒ $\overbrace{u(0.2,t)}^{u_1} = -0.4$ & $\overbrace{u(-0.2,t)}^{u_2} = 0.4$

but

$u(0.2,t)$ & $u(-0.2,t)$ ⇒ -0.4 & 0.1

↳ $\underline{-0.3}$ Symm.

$\neq 0$ Violation